



CoBuild 1.0 Hackathon



Team Name — **Coders**

TEAM MEMBERS



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Team Leader



Simran

Member



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Member



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Member

Problem Statement: PharmSecure

Two connected crises — one broken trust chain.

PROBLEM 1

Counterfeit Medicines

Fake or duplicated medicines can enter the supply chain. Patients and small clinics have no easy way to check if a medicine is genuine, authorized, or already been scanned elsewhere — putting patient safety directly at risk.

PROBLEM 2

Medicine Wastage

Hospitals often discard medicines still 30–60 days from expiry simply because they no longer need them — while nearby clinics or charities face shortages of those exact same medicines, with no system connecting the two.

THE COMBINED IMPACT

Genuine medicine gets wasted in one place while patients go without it in another — and there's still no reliable way to confirm what's authentic. PharmSecure solves both: verifying medicines and redistributing surplus before it's too late.

 **3.2%**

CDSCO-confirmed spurious/
substandard samples (2023–24)

 **20%**

Estimated fake medicine rate
in India — ASSOCHAM / USTR

 **30–60**

Days of usable shelf life
discarded as “surplus”

 **1M+**

Deaths linked to fake
medicines worldwide — WHO

Proposed Solution : PharmSecure

One Intelligence Layer. Two Critical Decisions.

- **Authenticity Intelligence**

Analyzes medicine identity, scan history, location and supply-chain behavior to detect anomalies.

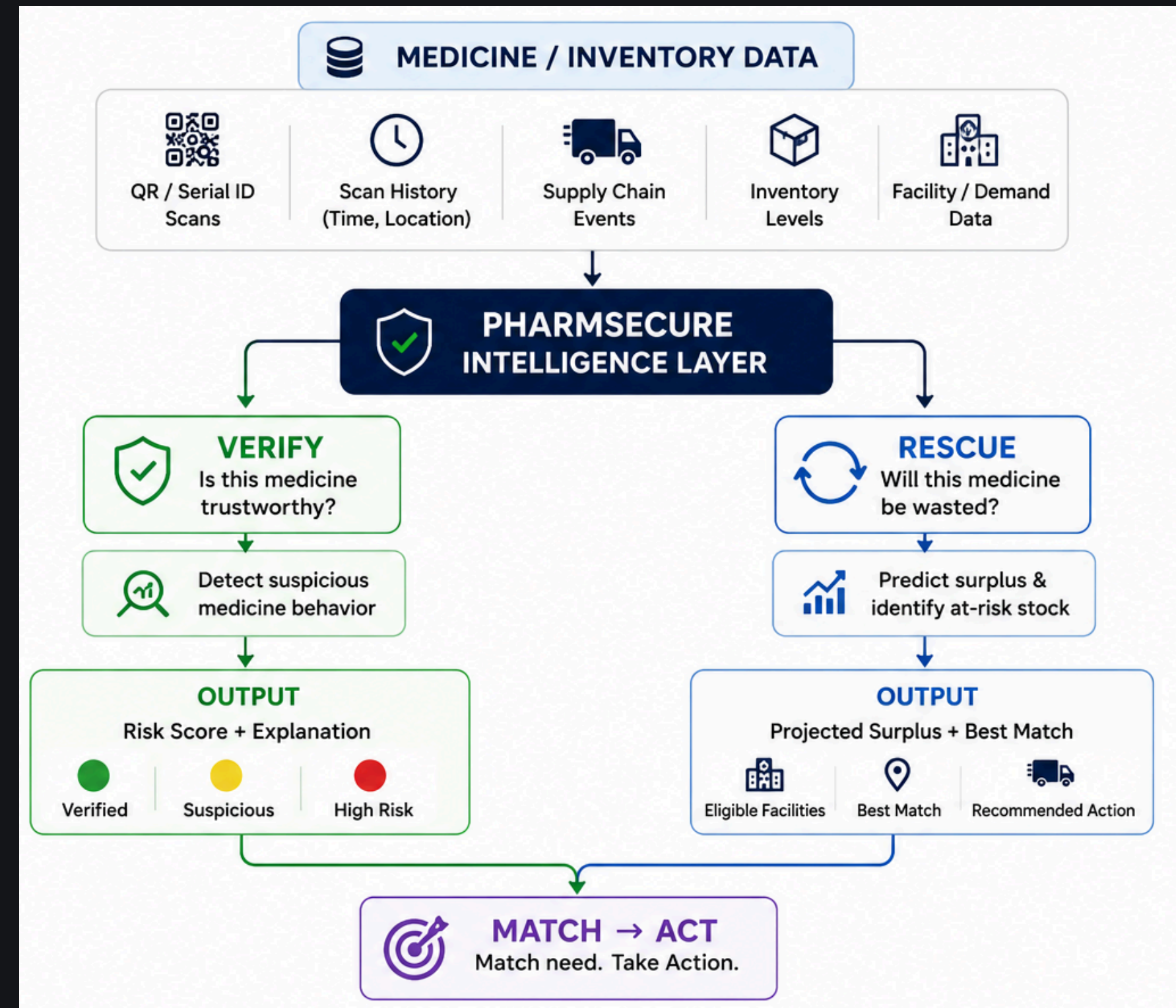
- **Expiry Intelligence**

Forecasts consumption and identifies inventory likely to become surplus before expiry.

- **Smart Redistribution**

Matches viable surplus with eligible facilities based on demand, distance, storage and expiry constraints.

Detect → Predict → Match → Act



PharmSecure Workflow

| FROM SIGNAL → DECISION → ACTION |



VERIFY

Can we **trust** this medicine?



PS-928174

Scanned



Delhi

10:02 AM



12 min



Jaipur

10:14 AM



Risk Score

91 / 100

HIGH RISK



Why flagged?



Duplicate identity



Geographic anomaly



Impossible travel interval



RESCUE

Will this medicine be **wasted**?

Inventory

Current 200

Forecasted use – 35

Safety stock – 20

Projected surplus **145**



145 units
projected surplus



47 days
remaining



Best match



Clinic B
120 units • 18 km



Hospital C
10 units • 8 km



Clinic D
60 units • 42 km



Recommended



120 units
to Clinic B

Hospital A 200 → **80**

Clinic B 0 → **120**

Tech Stack

Every component we use, and exactly how a request moves through the system.

STACK COMPONENTS

 **Frontend** React.js + Tailwind CSS
Dashboard, scanner & analytics UI

 **Backend** Node.js + Express.js
REST API, business logic

 **Database** PostgreSQL
Relational store, ACID transactions

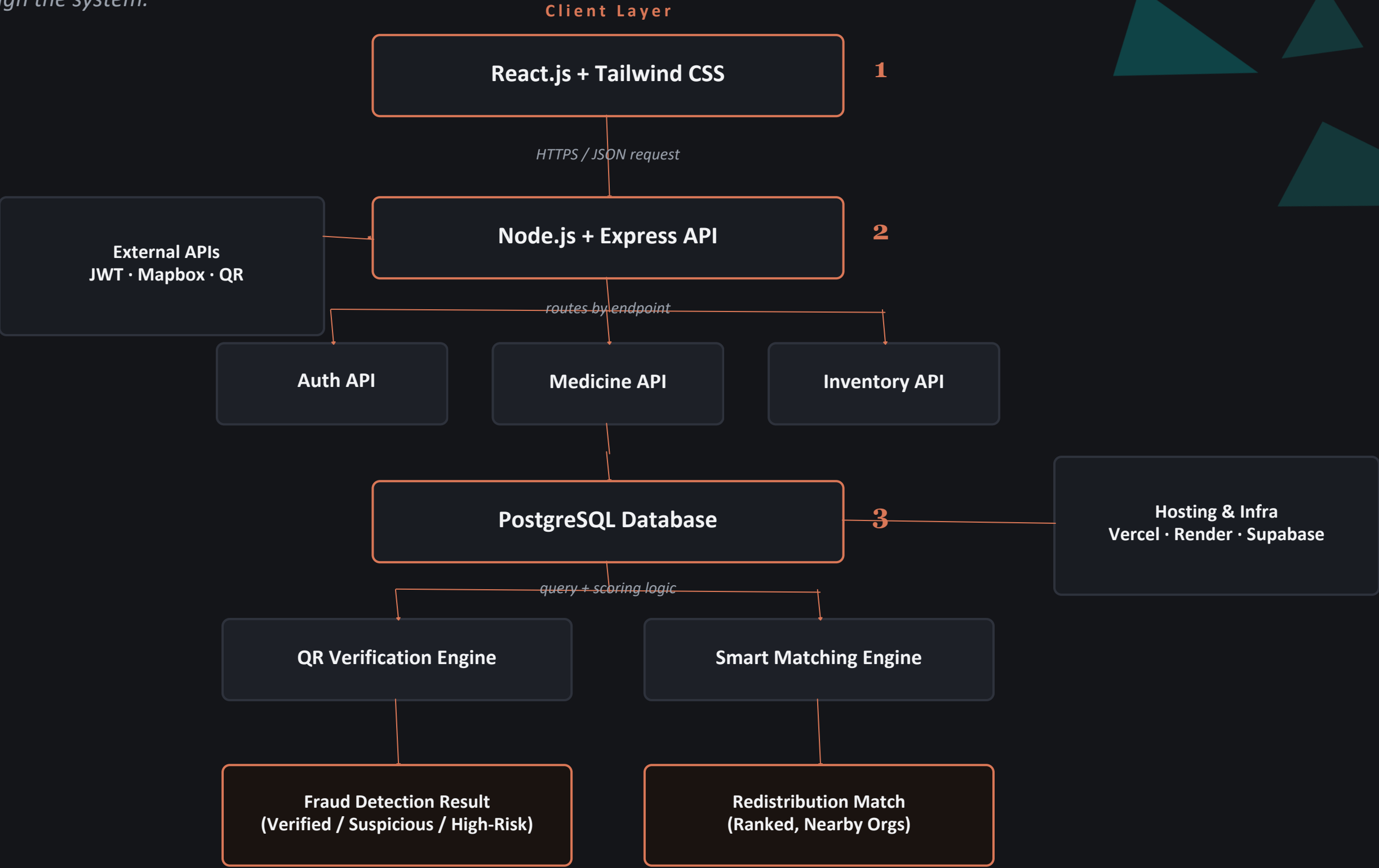
 **Intelligence** Risk Scoring + Matching
Explainable, weighted formulas

 **Auth** JWT
Stateless, role-based sessions

 **Maps** Mapbox GL
Live distance & location data

 **QR System** QR Code Library
Unique batch identity generation

 **Hosting** Vercel · Render · Supabase
Production deployment targets



↑ Response streams back to the Client — green check, red alert, or a matched transfer card

 Core pipeline stage

 Sub-service / route

 Final output

Key Platform features

Four systems working off one live inventory ledger.



Predictive Surplus Risk Score

Automated formula —
 $\text{Stock} - (\text{Daily Burn} \times \text{Days to Expiry}) - \text{Safety Buffer}$ —
surfaces expiring surplus in real time.



Smart Proximity Matching

Geospatial engine
balancing recipient
urgency, distance, transit
time, and historical
consumption capacity.



Cloned Code Anomaly Detector

Instantly flags duplicate
scans or geo-mismatches
— e.g. one serial scanned
in two cities within
minutes.

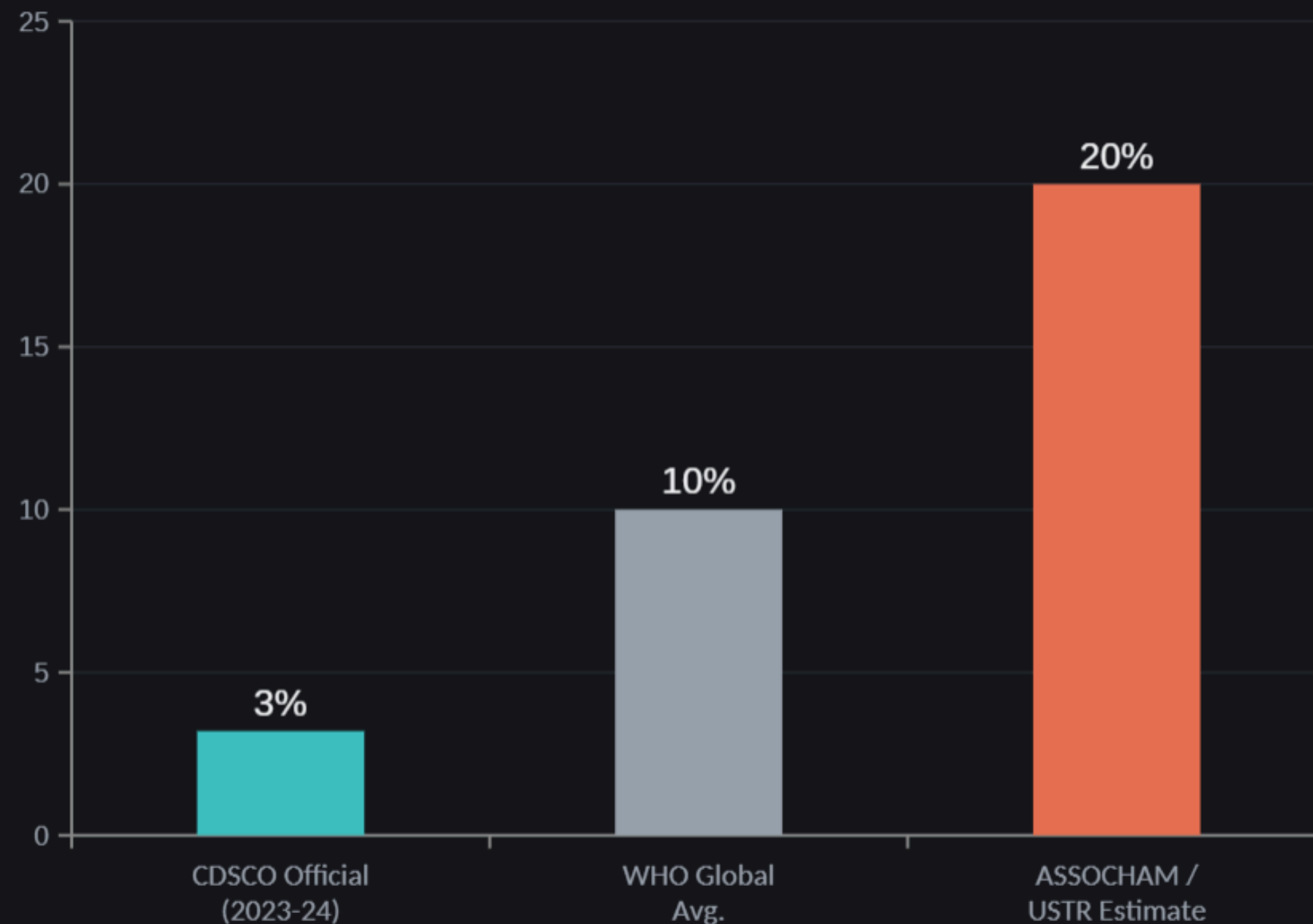


Audit Trail & Transfer Protocols

Generates compliance-
ready digital transfer logs,
preserving chain-of-
custody for every
redistribution.

Fake & Substandard Medicines in India — 2025 Data

Government figures vs. independent estimates — the gap itself is the problem.



OFFICIAL DATA

CDSCO's 2023–24 report found 3.2% of ~5,500 tested samples substandard or spurious. In July 2025 alone, 143 of the drugs checked failed quality standards, with 8 confirmed spurious.

THE WIDER ESTIMATE

WHO estimates ~10% of medicines globally are fake, causing over a million deaths a year. Industry bodies like ASSOCHAM and the US Trade Representative put India's figure as high as 20–25%.

PATIENT IMPACT

2025 CDSCO alerts flagged eye drops and IV fluids linked to blindness and maternal deaths in Karnataka — real harm from drugs that passed through the supply chain undetected.

Innovation & Impact

What makes PharmSecure different — and who it actually helps.



Innovation

1

Dual-Engine Platform

Combines authenticity verification and surplus redistribution in one system — most solutions solve only one.

2

Explainable Risk Scoring

Duplicate-scan and geo-anomaly detection flag fraud in real time, with a transparent score, not a black box.

3

Emergency Broadcast Matching

Clinics can request a medicine “now” and get instantly ranked, nearby matches — not just scheduled transfers.

4

FEFO-Driven Smart Matching

Weighted, explainable scoring (distance + urgency + demand + trust) any judge or hospital admin can audit.



Impact

1

Safer Patients

Instant verification reduces the chance of a counterfeit medicine reaching someone who needs it.

2

Less Wastage, More Savings

Near-expiry surplus gets redistributed instead of destroyed — recovering value hospitals currently write off.

3

Equity for Rural Clinics & NGOs

Smaller providers gain access to medicines they'd otherwise struggle to source or afford.

4

A Trust Network That Compounds

Every new hospital or clinic onboarded makes verification and matching stronger for everyone else.

Why it matters: PharmSecure doesn't just detect fake medicine — it prevents genuine medicine from becoming waste.

Feasibility

A realistic, low-risk path from hackathon prototype to real-world deployment.



PHASE 1

Prototype

NOW

This hackathon — 0–2 weeks

LOW EFFORT

- QR generation & scanning
- Fraud risk scoring
- Inventory & expiry alerts
- Redistribution matching
- Dashboard & analytics



PHASE 2

Pilot

NEXT

3–6 months, real orgs

MEDIUM EFFORT

- Onboard hospitals & clinics
- Verified organization accounts
- Live QR batches from manufacturers
- Trust score from real transfers



PHASE 3

Real Deployment

FUTURE

6–12+ months, at scale

HIGH EFFORT

- Hospital management system integration
- Direct manufacturer connections
- Government / regulatory integration
- Logistics partners for physical transfer



Technical

React + Node + PostgreSQL — well-documented, widely used stack.



Team & Resources

Buildable by a small 4-person team within a hackathon timeframe.



Cost to Run

Free-tier hosting (Vercel, Supabase) covers the pilot stage easily.



Regulatory Path

Starts as a verification aid, not a medical device — lower approval burden.

Bottom line: No blockchain, no custom AI models, no unavailable hardware — Phase 1 is demo-ready today.